

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Semester : V

Branch : Electrical and Power/Electrical & Electronics Engineering/Electrical Engineering

Subject Code & Name: Microprocessor and Microcontroller (BTEEC502)

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hr

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)	CO 01-05	12
1	Which of the following flag is used to mask INTR interrupt? a. zero flag b. auxiliary carry flag c. interrupt flag d. sign flag		1
2	Which of the following flag condition is used for BCD arithmetic operations in microprocessor? a. Sign flag b. Auxiliary carry flag c. Parity flag d. Zero flag		1
3	Which of the following is the correct sequence of operations in a microprocessor? a. Opcode fetch, memory read, memory write, I/O read, I/O write b. Opcode fetch, memory write, memory read, I/O read, I/O write c. I/O read, opcode fetch, memory read, memory write, I/O write d. I/O read, opcode fetch, memory write, memory read, I/O write		1
4	Which process is used to map logical addresses of variable length onto physical memory? a. Paging b. Overlays c. Segmentation d. Paging with segmentation		1
5	RAM is _____ and _____. a. Volatile, temporary b. Non-volatile, temporary c. Volatile, permanent d. Non-volatile, permanent		1
6	Which is of the following is true about SPHL instruction? a. It uses indexed addressing mode b. It is a 3-byte instruction c. It requires three T-states d. Contents of HL pair is moved to SP		1
7	8051 series has how many 16 bit registers? a. 2 b. 3 c. 1 d. 0		1

8	If more than one channel requests service simultaneously, the transfer will occur as				51646866	1
	a. Multi transfer	b. Simultaneous transfer	c. Burst transfer	d. None of the mentioned		
9	The bus is available when the DMA controller receives the signal					1
	a. HRQ	b. HLDA	c. DACK	d. All of the mentioned		
10	The 8255 is a _____ chip.					1
	a. Input /Output	b. Analog to Digital	c. Digital to analog	d. None of the mentioned		
11	How many rows and columns are present in a 16*2 alphanumeric LCD?				1	
	a. rows=2, columns=32	b. rows=16, columns=2	c. rows=16, columns=16	d. rows=2, column=16		
12	How are the status of the carry, auxiliary carry and parity flag affected if the write instruction MOV A,#9C ADD A,#64H				1	
	a.CY=0,AC=0,P=0	b. CY=1,AC=1,P=0	c.CY=0,AC=1,P=0	d.CY=1,AC=1,P=1		
Q. 2	Solve the following.				CO 1	12
A)	What is an interrupt? Explain a different type of interrupt occurs in 8085.					6
B)	Draw architecture diagram of 8085 microprocessor. Explain any two block.					6
Q. 3	Solve the following.				CO 2	12
A)	Discuss different types of data transfer technique in 8085.Explain synchronous and Asynchronous modes in 8085 microcontroller.					6
B)	Explain : 1.Memory mapped I/O Schemes 2.I/O mapped I/O schemes					6
Q. 4	Solve Any Two of the following.				CO 3	12
A)	Write short note on Stepper Motor.					6
B)	Explain DMA Controller.					6
	Explain traffic light controller in detail.				6	
Q. 5	Solve Any Two of the following.				CO 4	12
A)	Write short note on memory organization of 8051.					6
B)	Explain following instruction of 8051: 1.MOV A,#55H 2.ADD A,R2 3.ORG 0H					6
C)	Explain TCON & TMOD register in 8051 microcontroller.				6	

Q. 6	Solve Any Two of the following.	CO 5	12
A)	Write an assembly language program of 8051 for adding ten numbers in an array. assume suitable data.		6
B)	Draw interfacing diagram of 8k*8 program ROM with 8051 and also write memory map for same.		6
C)	Explain RS 232 and RS 485.		6
*** End ***			